READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

Effect and Cause

A chance finding by a Swiss research team explains an event that happened centuries ago around Lake Geneva

In the sixth century Gregory of Tours, a *chronicler** of the Germanic people known as the Franks, told of an extraordinary event in what is now Switzerland, where the Rhone River spills into Lake Geneva. He wrote of a big rockfall in the year 563 AD in the vicinity of a place called Tauredunum. The debris plunged into the river, and a great mass of water 'overwhelmed with a sudden and violent flood all that was on the banks as far as the city of Geneva,' more than 64 kilometres across the lake. The Geneva bridge was demolished, and several people inside the city walls of Geneva were killed. Historians and scientists have long believed that Gregory and another chronicler, Marius of Avenches, who told a similar tale, were describing a tsunami that raced across the lake. But there has not been any direct evidence of it until now.

Researchers at the University of Geneva now say they have found that evidence, in the form of a large deposit of sediment in the middle of the lake. In a study published in the journal Nature Geoscience, they also propose the sequence of events that caused the deadly surge. The researchers think that large boulders crashed down onto soft sediments which had accumulated at the river mouth because of the slowing of the river's flow when it enters the lake. These sediments formed an underwater delta that had several canyon-like channels. When the falling rocks hit the delta they destabilised the sediments and caused the canyons to collapse. It was this collapse that created the tsunami. The sediments from this collapse would have been propelled towards the lake's centre.

Guy Simpson, a lecturer in the University of Geneva's Department of Geology and Paleontology, says the thick layer of sediment, which has the same curved shape as a lens, lies more than 305 metres down in the deepest part of the lake, and was found largely by chance. Katrina Kremer, a University of Geneva doctoral student and the study's lead author, had been conducting seismic soundings, searching for thin sediment layers that might be evidence of major floods that had taken place in pre-historic times, long before the event described by Gregory of Tours. 'But we came across this enormous deposit,' Simpson says. 'We didn't know straight away that it was the deposit that caused the [sixth-century] tsunami. But it was a jumbled mess of sediment. It was quite obvious that it was deposited rapidly.'

**Chronicler: a person who writes an account of important or historical events*

The researchers then took samples of the sediments and used carbon-dating techniques on remains of leaves and other organic matter they found to determine when the deposit formed. This narrowed the range to a period between the late fourth century and the early seventh century. Other than the rockfall, there is no record of any special event during that period, Simpson says.

The researchers estimated that the deposit, which is at least 9.6 kilometres long by 4.8 kilometres wide, and averages about five metres thick, contains more than 248 million cubic metres of material. They ran multiple computer simulations showing that the collapse of that much sediment at the mouth of the Rhone would have caused a tsunami with an estimated height of 7.9 metres at Geneva - where it would have arrived in about 70 minutes. The rockfall itself may have been set off by a major earthquake, as some scientists have speculated.

Lake tsunamis, although unusual, are not unknown, says Richard Schweickert, an Emeritus Professor of Geology at the University of Nevada in Reno, in the United States. He cites evidence that the collapse of part of the shoreline of Lake Tahoe in northern California within the past 20,000 years caused a tsunami with wave heights of about 30 metres. There are two faults under the lake that could have caused an earthquake, he says and that the collapse of the Rhone delta sediments, as calculated by the Swiss researchers, 'would certainly be capable of moving a large amount of material into the lake.' He suggests that the findings could be corroborated by careful mapping of the shoreline to look for unusual deposits or erosion left behind by the giant waves.

Simpson says the Rhone delta sediments might collapse again, perhaps from an earthquake or even their own weight. In the sixth century, Geneva was a small community, mostly behind walls on a hill, whereas today it is home to international organisations and about 200,000 people, many living in low-lying areas near the water. Testing the stability of nearby slopes, and creating more detailed models of how a tsunami could affect Geneva today, would provide a more accurate assessment of whether or not this is something the lakeside city should be concerned about.

Most tsunamis occur in oceans and are generated by earthquakes. However, the study is a reminder that even a landlocked nation like Switzerland is not immune to catastrophic waves.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–7 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

1 According to Gregory of Tours, the landslide which caused the flood happened near Tauredunum.

2 The city of Geneva was undamaged by the tsunami that Gregory described.

3 The work of Marius of Avenches supported the idea that there was a tsunami.

4 The rocks which fell into the delta were very hard and dense.

5 Richard Schweickert has published studies on lake tsunamis that have occurred in several countries.

6 The shoreline of Lake Tahoe has remained unchanged for 20,000 years.

7 Parts of the population of Geneva now live closer to the lake than was the case in the sixth century.

Questions 8–13

Complete the flow-chart below.

*Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.*

Write your answers in boxes 8-13 on your answer sheet.

The evidence for a tsunami in Lake Geneva

Katrina Kremer was looking for indications of prehistoric **8** ……. that had occurred before the sixth-century event.

↓

Instead her research team found a large lens-shaped deposit.

↓

Researchers extracted samples from the deposit and dated organic material such as **9** ……. .

↓

Carbon-dating confirmed that the deposit was present in the lake by the beginning of the **10** ……. century.

↓

The volume of the deposit was then **11** ……. by the scientists.

↓

The researchers established the approximate **12** ……. and speed of the tsunami through extensive computer modelling.

↓

A likely cause of the rockfall was a significant **13** ……. .

Questions 1–7 TRUE / FALSE / NOT GIVEN

题号	答案	题干翻译	详细定位（第 X 段 + 定位句原文 + 定位句翻译）	详细解释
1	**TRUE**	根据 Gregory of Tours 的说法，引发洪水的山体滑坡发生在 Tauredunum 附近。	**第 1 段**：“*He wrote of a big rockfall in the year 563 AD in the vicinity of a place called Tauredunum.*”；翻译：他写到公元 563 年在一个叫 Tauredunum 的地方附近发生了一次大型岩崩。	题干说 “According to Gregory… happened near Tauredunum（发生在 Tauredunum 附近）”。原文明确说 **in the vicinity of… Tauredunum**，完全对应，因此为 **TRUE**。
2	**FALSE**	Gregory 描述的海啸（巨浪）并未对日内瓦造成破坏。	**第 1 段**：“*The Geneva bridge was demolished, and several people inside the city walls of Geneva were killed.*”；翻译：日内瓦的桥被摧毁，且日内瓦城墙内有数人死亡。	题干说 “undamaged（毫发无损）”。但原文给出两类明确破坏：**桥被摧毁 + 城内有人死亡**。这与 “未受损” 直接矛盾，所以 **FALSE**。
3	**TRUE**	Marius of Avenches 的记载支持 “确实发生过海啸” 的观点。	**第 1 段**：“*Historians and scientists have long believed that Gregory and another chronicler, Marius of Avenches, who told a similar tale, were describing a tsunami…*”；翻译：历史学家和科学家长期认为 Gregory 以及另一位编年史作者 Marius of Avenches（他也讲了一个相似的故事）描述的是一次海啸。	题干关键词 **supported the idea（支持这一观点）**。原文说 Marius **told a similar tale（讲了相似故事）**，并且学者据此长期相信他们在描述海啸——也就是把 Marius 的记载当作 “支撑/佐证”。因此 **TRUE**。
4	**NOT GIVEN**	落入三角洲的岩石非常坚硬且致密。	**第 2 段**：“*The researchers think that large boulders crashed down onto soft sediments…*”；翻译：研究者认为巨大的岩块砸到柔软的沉积物上……	文中只对 “被砸的东西” 给了性质：**soft sediments（柔软沉积物）**；对 **boulders（巨石/岩块）** 并没有说 “hard and dense（坚硬且致密）”。“boulder 常识上硬” 不能当成原文信息；题干两项性质（hard + dense）都未被明确陈述，所以 **NOT GIVEN**。
5	**NOT GIVEN**	Richard Schweickert 发表过关于多个国家湖泊海啸的研究。	**第 6 段**：“*Lake tsunamis… are not unknown, says Richard Schweickert…*”；“*He cites evidence that…*”；翻译：……Schweickert 说湖泊海啸并非闻所未闻；他引用证据表明……	原文只说明他是教授、并且 **cites evidence（引用证据）** 举了 Lake Tahoe 的例子；没有说他 **发表研究（published studies）**，也没有 “several countries（多个国家）”。信息缺失，因此 **NOT GIVEN**。
6	**FALSE**	太浩湖的湖岸线在两万年里一直没有变化。	**第 6 段**：“*He cites evidence that the collapse of part of the shoreline of Lake Tahoe… within the past 20,000 years…*”；翻译：他引用证据称，在过去两万年内，太浩湖部分湖岸线曾发生坍塌……	题干说 “remained unchanged（一直没变）”，但原文说两万年内 **发生过湖岸坍塌（collapse of part of the shoreline）**，这就意味着 “有变化”。与题干相反，故 **FALSE**。
7	**TRUE**	如今日内瓦的部分居民住得比六世纪时更靠近湖边。	**第 7 段**：“*In the sixth century, Geneva was a small community, mostly behind walls on a hill, whereas today… many living in low-lying areas near the water.*”；翻译：六世纪时日内瓦多在山上的城墙内；而如今很多人住在靠近水边的低洼地区。	对比结构 **whereas（而如今）** 很清晰：过去 “在山上城墙内”，现在 “低洼靠水”。这就是 “现在更靠近湖”。因此 **TRUE**。

Questions 8–13 Flow-chart（ONE WORD AND/OR A NUMBER）

题号	答案	题干翻译	详细定位（第 X 段 + 定位句原文 + 定位句翻译）	详细解释
8	**floods**	Katrina Kremer 在寻找史前 _____ 的迹象，这些事件发生在六世纪事件之前。	**第 3 段**：“*...searching for thin sediment layers that might be evidence of major floods that had taken place in pre-historic times...*”；翻译：.....寻找可能证明史前时期发生过重大洪水的薄沉积层。	空前是 “prehistoric（史前）”，原文对应名词是 **major floods**。按要求 “一词”，填 **floods**。
9	**leaves**	研究人员从沉积物中取样，并对诸如 _____ 等有机物进行测年。	**第 4 段**：“*...used carbon-dating techniques on remains of leaves and other organic matter...*”；翻译：.....对树叶残留物及其他有机物进行碳测年。	题干 “organic material such as ...” 对应原文举例 **leaves**，直接抄词即可。
10	**seventh**	碳测年证实：到 _____ 世纪初，这一沉积物已经存在于湖中。	**第 4 段**：“*This narrowed the range to a period between the late fourth century and the early seventh century.*”；翻译：时间范围缩小到四世纪末至七世纪初之间。	流程图说 “by the beginning of the ___ century（到.....世纪初）”。原文给出上限 **early seventh century（七世纪初）**，因此填 **seventh**。
11	**estimated**	科学家随后对沉积物的体积进行了 _____。	**第 5 段**：“*The researchers estimated that the deposit... contains more than 248 million cubic metres of material.*”；翻译：研究者估算该沉积物含有超过 2.48 亿立方米物质。	题干被动结构 “was then ___ by the scientists”，原文动词就是 **estimated（估算）**，语法与流程完全匹配。
12	**height**	研究人员通过大量计算机建模，确定了海啸的大致 _____ 和传播速度。	**第 5 段**：“*...caused a tsunami with an estimated height of 7.9 metres at Geneva...*”；翻译：.....在日内瓦处海啸估计高度为 7.9 米。	“approximate ___ and speed” 对应模型得出的关键指标之一是 **height（高度）**（并伴随到达时间用于推断速度）。填 **height**。
13	**earthquake**	岩崩的一个可能原因是一次显著的 _____。	**第 5 段**：“*The rockfall itself may have been set off by a major earthquake...*”；翻译：岩崩本身可能由一次大型地震触发。	“significant” 对应原文 “major”；触发原因明确是 **earthquake**，填词即可。

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